

PHASE 3 – SYSTEM DESIGN & ARCHITECTURE

Project: LegalEase – AI-Powered Legal Document Generator

1. Frontend Design

The Streamlit interface provides inputs for document type, parties, terms, and effective date. It also provides Generate Document, Edit Document, and download options.

2. Backend Design

FastAPI provides the backend API. The /generate endpoint accepts document_type, parties, terms, and dates and routes the request to the document-generation component.

3. AI Integration Design

GeminiDocumentGenerator forms a structured prompt and sends it to Gemini using generate_content. The generated text is returned to the backend and displayed by the frontend.

4. Document Formatting Design

DOCX output uses python-docx for titles, paragraph formatting, logo, terms tables, and footer. PDF output uses FPDF with headers, footers, logo, headings, and terms. HTML preview provides a styled on-screen representation.

5. User Flow

User enters document details → selects Generate Document → FastAPI receives the request → Gemini generates content → frontend displays the preview → user edits if required → user downloads TXT, DOCX, or PDF.

6. Design Goal

The design focuses on user interaction, AI-powered generation, editable previews, professional formatting, and multiple export formats.

Source basis: LegalEase project documentation supplied for this task.